**Unit:** undefined
**Topic:** Algebra 1: Polynomials

**Name:** ____________
**Date:** __________

# Level 2: Foundation

**Q1.** What is the degree of the polynomial $3x^2 + 2x - 4$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

**Q2.** Which of the following is a polynomial?
(A) $x^2 + 3x = 10$
(B) $x^2 + 3x + 1$
(C) $\frac{1}{x} + 2x^2$
(D) $x^3 - 2x^2 + x + 1$
(E) $\sqrt{x} + 2x^2$

**Q3.** What is the coefficient of $x$ in the polynomial $2x^2 + 5x - 3$?
(A) 2
(B) 5
(C) 3
(D) 1
(E) 0

**Q4.** Simplify the expression: $2x^2 + 3x + 4x^2$
(A) $6x^2 + 3x$
(B) $5x^2 + 3x$
(C) $2x^2 + 7x$
(D) $6x^2 + 7x$
(E) $5x^2 + 5x$

**Q5.** Factor out the greatest common factor from $6x^2 + 12x$
(A) $6x(x + 2)$
(B) $6x^2(2x + 1)$
(C) $6x^2(x + 2)$
(D) $6x(x + 1)$
(E) $3(2x^2 + 4x)$

**Q6.** What is the value of $x$ that makes the polynomial $x^2+4x+4$ equal to zero?
(A) -4
(B) -3
(C) -2
(D) 0
(E) -1

**Q7.** Which one of the following polynomials has a degree of 3?
(A) $2x^2 + x - 3$
(B) $3x^3 - 2x^2 + x - 1$
(C) $x^2 + 3x - 4$
(D) $5x - 2$
(E) $x^4 + 2x^2 - 1$

**Q8.** What is the result of adding $2x^2 + 3x - 4$ and $x^2 - 2x - 1$?
(A) $3x^2 + x - 3$
(B) $3x^2 + x - 5$
(C) $x^2 + x - 5$
(D) $3x^2 + 5x - 3$
(E) $x^2 - x - 3$

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**Open Ended Questions (FRQ)**

**Q9.** Draw a diagram to show how the polynomial $x^2 + 2x + 1$ can be factored as $(x + 1)^2$. Explain why this is a special factoring case.

**Q10.** Explain why $2x^2 + 5x - 3$ is a polynomial, but $\frac{1}{x} + 2x^2$ is not. Include definitions and examples to support your explanation.

**Q11.** Simplify the expression: $(2x^2 + 3x) + (x^2 - 2x)$. Draw a step-by-step solution and explain the reasoning behind combining like terms.

**Q12.** Factor out the greatest common factor from the expression $12x^3 + 18x^2$. Show your work and explain why this is an important step in simplifying polynomials.

**Chef's Tips**



- Emphasize the importance of understanding the definition of a polynomial and how to identify the degree, terms, and coefficients.
- Practice combining like terms and factoring out the greatest common factor to simplify polynomials.
- Use visual aids and diagrams to help students understand the relationship between polynomials and their factored forms.